$$\left.\begin{aligned} X_a &= \frac{1}{2}v_0 t + \frac{v_0}{2\omega_0}\sin\omega_0 t - \frac{l}{2} \\ X_b &= \frac{1}{2}v_0 t - \frac{v_0}{2\omega_0}\sin\omega_0 t + \frac{l}{2} \\ X_c &= \frac{1}{2}v_0 t + \frac{v_0}{2\omega_0}\sin\omega_0 t + \frac{l}{2} \\ X_d &= \frac{1}{2}v_0 t - \frac{v_0}{2\omega_0}\sin\omega_0 t + \frac{3l}{2} \end{aligned}\right\} \quad (7)$$

$$\left.\begin{aligned} v_a &= \frac{v_0}{2} + \frac{v_0}{2}\cos\omega_0 t \\ v_b &= \frac{v_0}{2} - \frac{v_0}{2}\cos\omega_0 t \\ v_c &= \frac{v_0}{2} + \frac{v_0}{2}\cos\omega_0 t \\ v_d &= \frac{v_0}{2} - \frac{v_0}{2}\cos\omega_0 t \end{aligned}\right\} \quad (8)$$

$$X_b = X_c$$

$$\sin\omega_0 t = 0 \quad (9)$$

$$\omega_0 t = \pi \quad (10)$$

$$v_b'(\ \ ) = \frac{v_0}{2}$$

$$v_c'(\ \ ) = -\frac{v_0}{2}$$

$$v_b'(\ \ ) = -\frac{v_0}{2}$$

$$v_c'(\ \ ) = \frac{v_0}{2}$$

$$x_a' = 0$$

$$x_b' = 0$$

$$x_c' = 0$$

$$x_d' = 0$$